Rohan Holikatti

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.41
R.S. Chandak High School & S.E.S. Jr. College of Arts & Science	Class XII (Maharashtra State Board of Secondary and Higher Secondary Education, Pune)	2023	75.00%
Indian Model School, Akkalkot	Class X (Maharashtra State Board of Secondary and Higher Secondary Education, Pune)	2021	94.40%

WORK EXPERIENCE

Quant Intern, Futures First

- Built an interactive **meeting-premia pricing tool** in **Python (Pandas, NumPy, ipywidgets)** that converts **RBA and Fed Funds futures** prices into per-meeting rate expectations (**CME FedWatch** method), with a curve comparison module across two timestamps.
- Analysed **5M+ ten-minute futures bars** using **Statsmodels and SciPy**: seasonality, **ADF, Hurst, ACF/PACF**, cointegrated FF-SRA spread, **PCA**, lead-lag, and anomaly and regime flags.
- Built a vectorised **backtesting engine** in **Python** for **z-score mean-reversion** on spreads and butterflies in US and Australian markets.
- Coded an **IB-implied OIS path** and measured repricing across **19 RBA decisions**, plus an **RBA statement comparison tool** that tags hawkish/dovish changes and updates meeting premia.
- Ran a **60:40 portfolio study** with monthly rebalancing and a **SciPy** Sharpe-optimised 4-asset blend.
- Developed **Streamlit and Plotly dashboards**: a 4-country macro dashboard (US, AUS, JPN, NZ) and an interactive futures candlestick dashboard with central bank and data-release overlays.

SKILLS

Programming Languages: Proficiency in C++, Python, HTML, CSS, MySQL

Python Libraries: Numpy, Pandas, Matplotlib, Scikit-learn, Tensorflow, Seaborn

Tools: Excel, PowerPoint, Word, VS Code, Jupyter Notebook, Git, GitHub, Google Colab, Railway, Vercel

Quantitative Finance: STIR Futures Pricing, Central Bank Meeting Premia, OIS Path Construction, Central Bank Communication Analysis, Event Studies, Scenario Analysis, Spread and Butterfly Strategies, Time Series Testing (ADF, ACF/PACF, Hurst), Seasonality Analysis, Backtesting, Portfolio Optimisation

RELEVANT COURSES

IT Courses: Introduction to Programming, Introduction to AI, Computer vision in Chemical And Life Science

Mathematics Courses: Elementary Linear Algebra, Applied Mathematics, Numerical Methods, Calculus-1, Calculus-2, Differential Equations

Machine Learning(Credential): Supervised Machine Learning: Regression and Classification Credential, Advanced Learning Algorithms by Andrew NG Credential, Unsupervised Learning, Recommenders, Reinforcement Learning Credential

PROJECTS

Fantasy Football Squad Selector:

- Built an end-to-end ML pipeline predicting Fantasy Premier League player points using **Random Forest** (selected over **Ridge Regression** and **XGBoost** via **backtesting** on historical Premier League data, achieving **MAE 1.02**).
- Engineered **25+ features** including rolling player form, team-level defensive strength, price-trend signals, and season-recency weighting; validated every design decision through backtesting rather than assumption.
- Designed a **linear programming optimizer (PuLP)** for squads, starting XIs, and transfers under budget/formation constraints, plus a **shrinkage-based Bayesian blending** to balance current vs. prior-season data.
- Built a live decision-support system integrating the official FPL API for real-time squad state, automated transfer recommendations, and chip-usage strategy calibrated via historical optimal-stopping analysis.

SPY Intraday Trading Strategy:

- Built an end-to-end quant data pipeline on **6+ years of 5-minute SPY data** (**118K+ bars**, via **Alpaca API**), with a custom **walk-forward backtesting engine** and **plug-in strategy interface** for rules-based and ML strategies.
- Validated market efficiency via **ADF, Hurst exponent, ACF/PACF, variance ratio**, then engineered **16 features** (return z-scores, realized volatility, volume, session-time) with strict no-lookahead guarantees.
- Walk-forward validated **Linear Regression**, then achieved **IC = 0.014(p<0.02)** and **Sharpe = 0.77** on a **30K+ bar test set**. Stress-tested via **Monte Carlo** and automated via **GitHub Actions** for automated signal generation (dry-run).

FPL Companion Dashboard:

- Built a full-stack fantasy football dashboard that syncs live data from the official FPL API, with **JWT authentication** with **bcrypt** password hashing, letting users securely link their FPL team and view live squad, points, and rank.
- Designed a **node-cron** scheduled pipeline auto-refreshing **650+ players** and **380 fixtures** every **6 hours**; built player comparison charts, fixture-difficulty heatmaps, and a form-and-fixture-based transfer-suggestion engine.
- Deployed backend, PostgreSQL on **Railway,** frontend on **Vercel**, with automatic migrations and CI-based deploys.

POSITIONS OF RESPONSIBILITY

PR & Outreach Core @ Finance and Consulting Club, IITH

Marketing Coordinator @ Cultural And Literary Council 2024-25

EXTRACURRICULAR

Active Member of NSS, IIT Hyderabad, contributing to campus-wide social service initiatives

Passed the Intermediate Grade Drawing Examination conducted by the Government of Maharashtra.